EXPERIMENT-12

12. Scenario: You are working on a data analysis project that involves analyzing the monthly temperature and rainfall data for a city. You have a dataset containing the monthly temperature and rainfall values for each month of a year. Your task is to develop a Python program that generates line plots and scatter plots to visualize the temperature and rainfall data.

Question:

1. Develop a Python program to create a line plot of the monthly temperature data.

2: Develop a Python program to create a scatter plot of the monthly rainfall data.

Question 12(a): Line Plot of Monthly Temperature

Aim: To visualize monthly temperature data using a line plot.

Algorithm:

1. Import Matplotlib.
2. Define months and temperature values.
3. Create a line plot using plt.plot().
4. Add labels and display the plot.

Code:

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']

temperature = [25, 27, 30, 33, 35, 32]

plt.plot(months, temperature, marker='o')

plt.xlabel("Month")

plt.ylabel("Temperature (°C)")
```

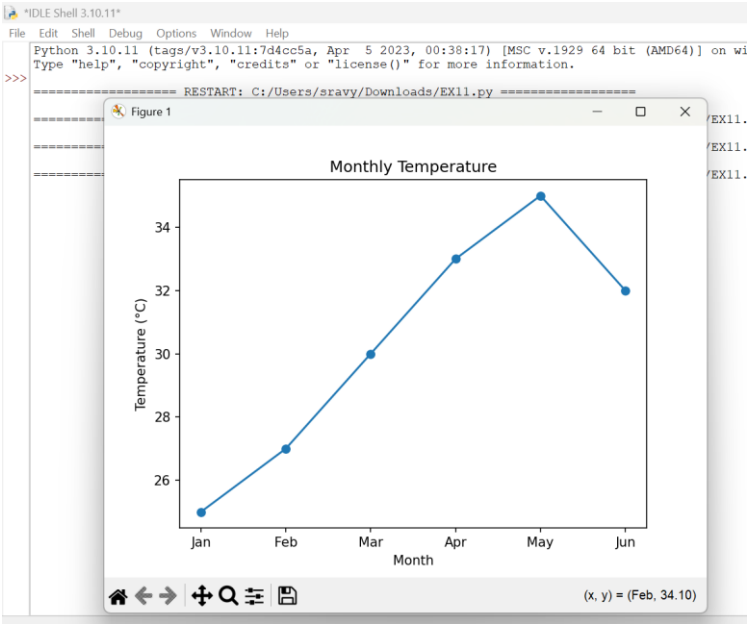
plt.title("Monthly Temperature")

plt.show()

Input:

Temperature = [25, 27, 30, 33, 35, 32]

Output: A line plot showing monthly temperatures, with May having the highest temperature.



Result: The monthly temperature data was visualized successfully using a line plot.

Question 12(b): Scatter Plot of Monthly Rainfall

Aim: To visualize monthly rainfall data using a scatter plot.

Algorithm:

1. Import Matplotlib.
2. Define months and rainfall values.
3. Create a scatter plot using plt.scatter().
4. Add labels and display the plot.

Code:

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']

rainfall = [20, 35, 50, 40, 80, 100]

plt.scatter(months, rainfall)

plt.xlabel("Month")

plt.ylabel("Rainfall (mm)")

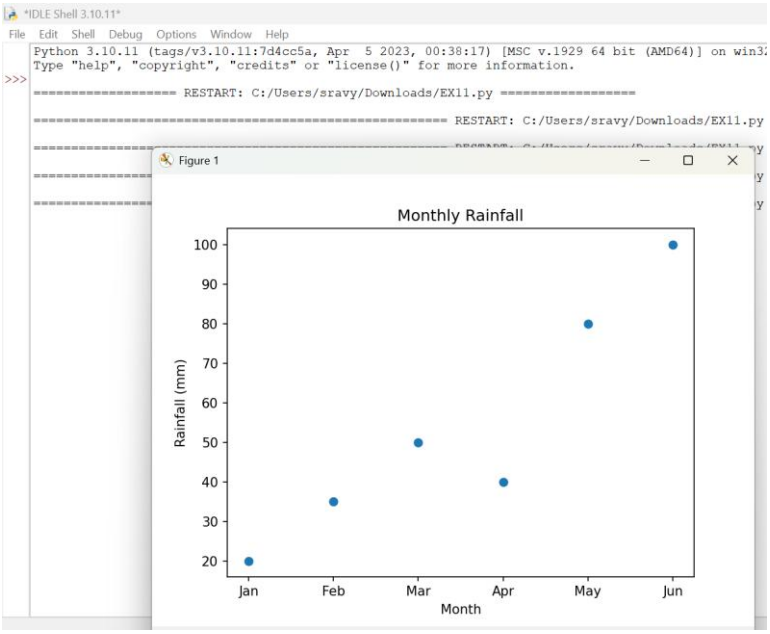
plt.title("Monthly Rainfall")

plt.show()
```

Input:

Rainfall = [20, 35, 50, 40, 80, 100]

Output: A scatter plot showing monthly rainfall, with June having the highest rainfall.



Result: The monthly rainfall data was visualized successfully using a scatter plot.